

Course Title:

Evaluation Methods for NLP

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Credits: 2

1.Prerequisite Course / Knowledge:

Introduction to NLP

Course Objectives

After completing this course, students will be able to:

1. Understand the importance of evaluation in NLP and the differences between automatic and human evaluation methods.
2. Apply various evaluation metrics to assess the performance of NLP models and algorithms.
3. Utilize human evaluation frameworks effectively to gather qualitative insights.
4. Address challenges in evaluating NLP models in multilingual and low-resource settings.
5. Benchmark NLP models using widely accepted datasets and evaluation protocols.

Syllabus

- **Unit 1: Introduction to Evaluation in NLP:** Covers the basics of evaluation, metrics (precision, recall, F1-score, BLEU, ROUGE), paradigms (intrinsic vs. extrinsic, task-based vs. corpus-based), and statistical significance testing.
- **Unit 2: Human Evaluation Techniques:** Discusses the role of human judgments, frameworks (crowd-sourcing, expert evaluation), quality annotation, and Multidimensional Quality Metrics (MQM).
- **Unit 3: Automatic Evaluation Techniques:** Focuses on metrics (BLEU, METEOR, TER, COMET, ROUGE, content-based metrics, precision, recall, F1-score).

- **Unit 4: Advanced Topics in NLP Evaluation:** Explores bias and fairness in evaluation, challenges in multilingual and low-resource settings, and benchmarking with GLUE, SuperGLUE, and IndicGLUE.

Detailed Syllabus:

Unit 1: Introduction to Evaluation in NLP

- **Class 1: Introduction to Evaluation in NLP**
 - Importance of evaluation in NLP
 - Overview of automatic vs. human evaluation methods
 - Evaluation metrics: Precision, Recall, F1-Score, Accuracy, BLEU, ROUGE
- **Class 2: Evaluation Paradigms**
 - Intrinsic vs. extrinsic evaluation
 - Task-based vs. corpus-based evaluation
 - Benchmarking datasets and quality analysis
- **Class 3: Statistical Significance Testing**
 - T-test, Chi-square, Bootstrapping
 - Confidence intervals and their importance in NLP evaluation

Unit 2: Human Evaluation Techniques

- **Class 4: Human Evaluation in NLP**
 - Role and importance of human judgments
 - Human evaluation frameworks: Crowd-sourcing, Expert evaluation
- **Class 5: Quality Annotation and Rating Scales**
 - Likert scales, Semantic Differential scales
 - Annotator agreement: Cohen's Kappa, Krippendorff's Alpha
- **Class 6: Multidimensional Quality Metrics (MQM)**
 - Introduction to MQM framework
 - Error categories and quality dimensions in MQM
 - Applying MQM in Machine Translation and other NLP tasks

Unit 3: Automatic Evaluation Techniques

- **Class 7: Machine Translation Evaluation**
 - BLEU, METEOR, TER, COMET
 - Challenges and limitations of automatic metrics

- **Class 8: Evaluation of Text Summarization**
 - ROUGE, Content-based metrics
 - Summary Length, Coherence, Relevance
- **Class 9: Evaluation of Text Classification and Information Retrieval**
 - Precision, Recall, F1-Score,
- **Class 10: Evaluating Embedding Quality**
 - Word Embedding benchmarks
 - Sentence embeddings and their evaluation

Unit 4: Advanced Topics in NLP Evaluation

- **Class 11: Bias and Fairness in NLP Evaluation**
 - Detecting and measuring bias in NLP models
 - Fairness-aware evaluation metrics
- **Class 12: Evaluation in Multilingual and Low-resource Settings**
 - Challenges in multilingual NLP evaluation
 - Transferability and generalizability of evaluation metrics
- **Class 13-14: Benchmarking with GLUE, SuperGLUE, indicGLUE and Beyond**
 - Introduction to GLUE and SuperGLUE benchmarks
 - Evaluating general language understanding with GLUE

Evaluation Methods for NLP

References:

- Jurafsky, D., & Martin, J. H. (2024). *Speech and Language Processing* (3rd ed.). Prentice-Hall.
- Manning, C. D., & Schütze, H. (1999). *Foundations of Statistical Natural Language Processing*. MIT Press.
- Gehrmann, S., Belz, A., & Berant, J. (2021). The GEM benchmark: Natural language generation, its evaluation, and metrics. *Transactions of the Association for Computational Linguistics*, 9, 165-184.
- Sellam, T., Das, D., & Parikh, A. (2020). BLEURT: Learning robust metrics for text generation. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 7881-7892.

- Leiter, J., Rodrigues, P., Kaushik, D., & Lipton, Z. C. (2022). Towards explainable evaluation metrics for natural language generation. *arXiv preprint arXiv:2205.01961*.
- Papinenko, A., Roumi, P., Turunen, D., & Kazakov, V. (2002). BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*. Philadelphia, Pennsylvania.
- Mathur, S., Freitag, A., Kocmi, A., & Liu, Y. (2020). COMET: A Neural Framework for MT Evaluation. arXiv preprint arXiv:2004.14040. **GitHub Repository:** <https://github.com/Unbabel/COMET>
- Lommel, A., Graham, D., & Popovic, M. (2014). Multidimensional Quality Metrics for Machine Translation: A Framework and Evaluation. In *Proceedings of the Ninth International Workshop on Statistical Machine Translation*.
- **GLUE (General Language Understanding Evaluation):** <https://gluebenchmark.com/>
- **SuperGLUE (Supernatural Language Understanding Evaluation):** <https://super.gluebenchmark.com/>
- **indicGLUE Benchmark:** https://huggingface.co/spaces/evaluate-metric/indic_glue

Teaching-Learning Strategies

This course combines theory-based and hands-on learning, with a focus on applying evaluation techniques to real-world NLP tasks. The approach includes:

- **Lectures:** to introduce and explain fundamental concepts of evaluation in NLP, including both automatic and human evaluation techniques.
- **Case Studies:** to analyze different evaluation metrics and their application to various NLP tasks like Machine Translation, Text Summarization, and more.
- **Practical Sessions:** to provide students with the opportunity to work with datasets and benchmarks, applying evaluation techniques to NLP models.
- **Projects and Group Discussions:** to foster collaborative learning and problem-solving skills, focusing on the challenges of multilingual and low-resource NLP evaluation.

Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO1	3	3	3	3	3	1	1	3	3	2	2
CO2	3	3	3	2	1	1	1	3	3	3	2
CO3	1	3	3	3	2	1	1	3	3	3	3
CO4	1	1	1	1	1	2	3	3	2	2	2
CO5	1	2	2	1	1	2	3	3	3	2	2

'3' in the box denotes 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

Assessment Methods and Weightages

This course emphasizes evaluating and benchmarking NLP models using both theoretical knowledge and practical applications. Students will gain hands-on experience by engaging with projects and case studies involving real-world datasets.

Type of Evaluation	Weightage (in %)
Assignments	30%
Seminar	10%+10% =20%
Project	40%-10% = 30%
Quiz	20%

Seminar Reading List: Evaluation Methods for NLP

<https://docs.google.com/spreadsheets/d/1oq-9k4dw9D84ZzTosgvEp-EOHNhiRJ4NLOUL3FA-oow/edit?usp=sharing>